

CALIMESA FIRE DEPARTMENT

POLICY MANUAL

VOLUME 6

Policy: 600.22 Response to Power Lines/Energized Electrical Equipment

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It is our policy to respond to reports of power lines down and other hazards involving energized electrical equipment (transformers, substations, electric vaults) for fire control and public safety. It is the responsibility of the company officer to maintain that level of safety until relieved by another fire company, police agency or utility company.

This guideline will establish a standard approach and response to the report of power lines down and other responses to energized electrical equipment. Power lines can come in contact with the ground as a result of storm related activity, fire, or vehicles striking power poles. In all cases, the potential for electrical shock/electrocution and secondary fire must be considered.

ELECTRIC SAFETY AWARENESS

Electricity will travel any conductive path it can as it seeks a ground. A direct path to ground can occur when contact is made between something energized and a portion of your body such as your hand, arm, head or other body part. An indirect path to ground occurs when you are holding something or touching an object that is in contact with something energized. This could include tools or other equipment you may be holding or when touching a fence, vehicle, or other object that may be in contact with something energized.

Gradient Voltage (Step and Touch Potential): When power lines are down, they will energize the ground around them. For Example: point of ground contact could be 7160 volts. This voltage will lessen as it radiates out from this point; for example, 6000 volts. If your feet are in areas where there is a voltage difference, you could complete the circuit and be the source to ground. This is called “step potential”. This danger could be indicated by a tingling sensation in the feet and serve as a warning to back away from the area. Step potential is more severe when the ground is wet.

KEY POINTS

1. Downed lines must always be considered energized with potentially lethal current.
2. Lines can reset and become “hot” or “energized” again by manual operation of a switch, by automatic re-closing methods (either method from short or long distances away), by induction where a de-energized line can become hot if it’s near an energized line or through back feed conditions.
3. Power line tends to have “Reel Memory” and may curl back or roll on itself when down.
4. Use caution when spraying water on or around energized electrical equipment. Hose streams conduct current! Never spray directly into the power lines. Use a fog spray at the base of the pole. Your primary responsibility is to protect the surrounding area. Short bursts of water are preferred methods to avoid being grounded. Never spray water onto electrical equipment until a utility rep has confirmed that the equipment is de-energized or “dead.”

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5. Electrical equipment is classified as:
 - a. Energized
 - b. De-energized (cannot be 100% guaranteed)
 - c. Dead (confirmed by utility representatives after grounding the lines(s).
6. PCB hazards: Smoke potentially fatal; avoid and contain pools of oil around transformers.
7. Poor soil resistance in the desert southwest may not provide enough of a ground to trip a circuit even when a conductor is laying on it.
8. You cannot tell the voltage of a power line by the size of the conductor. Most overhead conductors are not insulated
9. Voltage can travel through both dry and especially wet ground for considerable distances.
10. Pad-mounted and overhead transformers can explode.
11. Until grounded, equipment can contain electric potential, which can cause severe injury or death.
12. Electricity can flow through the ground or other conductive objects, (fences) to point far from the scene.

RESPONSE TO POWER LINES DOWN

1. Request utility company to respond.
2. Consider all down wires as “energized.”
3. Place apparatus away from “down lines and power poles” and out from under involved overhead lines that could fail and fall onto equipment or personnel.
4. Secure the area/deny entry.
5. Periods of high activity; company officer may choose to leave one (1) crewmember on-scene with a radio to wait for utility company.
6. In the event of multiple lines/poles down over a large area, call additional resources.

DOWN POWER LINES AND VEHICLES

1. Request utility company to respond.
2. Do not touch vehicle
3. Have occupants remain inside the vehicle
4. Place apparatus a safe distance away from down lines.
5. If occupants must leave the vehicle (fire or other threat to life) instruct them to open the door, not step-out! They should jump free of the vehicle without touching vehicle and ground at the same time; they should walk away from the vehicle with very small steps.

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SUB-STATION, TRANSFORMER, ELECTRICAL VAULT AND MANHOLE FIRE

1. Request utility company to respond.
2. Clear the area.
3. Be aware of explosion potential.
4. Place apparatus in a safe location away from overhead power lines.
5. Protect exposures.
6. Do not make entry until the utility representative has verified that the above electrical equipment has been de-energized. The utility representative may have to make entry to uninvolved sections to safely de-energize the equipment.

RESPONSE TO POWER POLE FIRES

1. Request utility company to respond.
2. Consider all wires and poles as “energized.”
3. Place apparatus away from “down lines and power poles” and out from under involved overhead lines that could fail and fall onto equipment or personnel.
4. Secure the area/deny entry.
5. Do not make any fire attack until the utility representative has verified that the electrical equipment has been de-energized.

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The determination of fire origin and cause is necessary for all fire incidents. The Incident Commander of the incident is responsible for ensuring a fire cause and origin investigations are conducted within the City of Calimesa jurisdiction.

The Incident Commander may elect to utilize personnel trained in fire origin and cause. If a more in depth investigation is necessary the Incident Commander shall make a request for a fire investigator who is proficient in the type of in depth fire investigation.

The following fire incident circumstances shall require a more in depth fire investigation:

- Fire deaths or serious fire injury
- All working greater structure fires
- Fires for which the cause cannot be determined by on scene resources
- Explosions and bombings
- Requested by law enforcement to respond to a known arson fire
- Car, field, or dumpster fires that have an identified lead or suspect in custody
- Attempted arson

FIRE SCENE COORDINATION

Fire origin and cause investigations shall be conducted after a fire is controlled but before overhaul actions occur to prevent or hinder of the investigation.

PRESERVATION OF EVIDENCE

Every fire scene contains evidence. Firefighting operations present the greatest potential for damage to evidence that may be used by the Fire Investigators and Law Enforcement in subsequent court cases and prosecution.

Evidence of fire cause is rarely destroyed by fire. The form, shape, color, size, and weight of items may be altered but evidence can still be identified upon trained examination. It is imperative to preserve evidence as found in place and not moving fire debris unnecessarily.

It is the responsibility of the Fire Department members to protect the fire scene from unnecessary damage during firefighting operations. Special care exercised during extinguishment will avoid the destruction of evidence through the misuse of fire streams.

Salvage operations should be minimal until the initial fire investigation is completed and should be confined to diminishing loss. Companies assigned to salvage operations should incorporate scene security and evidence preservation into their plan to stop the loss.

The fire scene is the Fire Investigator's laboratory. It is searched carefully and thoroughly

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photographed, diagrammed for placement of contents and evidence.

Evidence is then collected and preserved and the fire scene must be secured. Evidence cannot be used in court unless the Fire Investigator can establish a chain of custody by proving who found the evidence, where it was found and the evidence was not tampered with while in official custody. To ensure that the chain of custody remains unbroken the scene must remain in the sole custody of the Fire Department.

When at a fire scene where custody must be maintained a guard must be posted and custody must be maintained until the scene is released. No unauthorized persons may enter the scene. The Fire Department has the legal authority to close the scene entirely even to the property owner or to another interested person/s.

It is vital that the Fire Department prevent personnel from unnecessarily walking through a fire area, walking on, obscuring evidence, or picking up and moving evidence. This includes both Fire Department personnel and the media. If it is essential that evidence be moved or if necessary firefighting operations may damage evidence the evidence must be covered or its location marked before moving it carefully to a secure location.

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CALIMESA FIRE DEPARTMENT REHABILITATION POLICY

POLICY:

It is the policy of the Calimesa Fire Department to ensure the health, safety, and well-being of our employees during incidents and/or training assignments.

No employee shall be permitted to continue emergency operations or training exercises beyond safe levels of physical or mental endurance.

SCOPE:

The Incident Commander / Supervisor (Company Officer) shall consider the use of REHAB on any incident or training exercise where personnel are performing arduous or extensive work or exposed to high or low ambient temperatures.

RESPONSIBILITY:

- A. It shall be the responsibility of the Incident Commander / Supervisor to implement REHAB protocols.
- B. It is the responsibility of every individual to monitor his or her own exertion levels and capabilities. It is also the responsibility of every individual to comply with individual levels of adequate rest, nutrition, and hydration guidelines and recommendations.

PROCEDURE:

- A. A rehabilitation Group Supervisor (Radio Designation REHAB) will be implemented to evaluate personnel for the effects of physical and mental exertion during and incident or training exercise where this Policy is implemented.
- B. Incident Commanders shall order enough resources to effectively REHAB crews assigned to incidents and training.
- C. The REHAB Group will provide a designated area(s) where personnel will assemble for:
 - a. A physical assessment
 - b. Rest, hydration, and nutrition.
 - c. Medical evaluation & continuous monitoring
 - d. Reassignment

COMMAND STRUCTURE:

- A. REHAB is in the Service Branch of the Logistics Section alongside the medical Unit. On smaller incidents REHAB may work under the Operations Section or report directly to the IC. The extent of the incident dictates the location for REHAB within the command structure.